TEAM SPUTNIK · CubeSat Avionics & Telemetry

Mission Engineering Presentation Deck & Technical Defense

MISSION PROJECT ROSTER · TEAM NAME: TEAM SPUTNIK

PROJECT MEMBERS & FLIGHT CREW:

1. **UJJWAL YADAV [TEAM LEADER]**
2. **PRIYANSHU YADAV**
3. **PRAKHAR DHAR DUBEY**
4. **RATNESH KUMAR SINGH**
5. **ATUL KUMAR MAURYA**

EXECUTIVE OVERVIEW:

Welcome to the official CubeSat Telemetry & Avionics Presentation Deck engineered and presented by TEAM SPUTNIK. This comprehensive technical study encompasses 1U satellite bus design, ESP32 dual-core flight software, multi-sensor telemetry acquisition, and ground station operations.



CURRICULUM ARCHITECTURE & AGENDA

- Modules 01-07: CubeSat Form Factor, LEO Orbital Mechanics & Sensor Arrays
- Modules 08-11: Embedded Microcontrollers, ESP32 Architecture & Pinouts
- Modules 12-16: ADC Quantization, Steinhart-Hart Equations & Motion Gyroscopes
- Modules 17-22: Communication Protocols, Ground Station, NASA Open MCT & WebSocket Bridge

Equipped with 22 comprehensive technical modules, interactive telemetry instrumentation, formula derivations, and evaluators viva questions.

TEAM NAME	TEAM LEADER	ENGINEERING CADRE	SPACECRAFT STANDARD
TEAM SPUTNIK	Ujjwal Yadav	5 Members	1U CubeSat (10×10×10 cm)

What is a CubeSat? & Telemetry

Standardized Nanosatellite Architecture & The Lifeline of Spacecraft Data

CORE THESIS & DEFINITION:

A CubeSat is a miniaturized satellite standard based on 10×10×10 cm cubic units (1U). Telemetry is the continuous remote transmission of internal health, operational vitals, and payload sensor data to Earth.



What is a CubeSat? & Telemetry · Aerospace Technical Reference

The CubeSat Form Factor Standard

Established in 1999 by Prof. Jordi Puig-Suari (Cal Poly) and Prof. Bob Twiggs (Stanford). A standard 1U CubeSat measures exactly 10 cm × 10 cm × 10 cm with a nominal mass limit of ~1.33 kg to 2.0 kg. Modular scales include 1.5U, 2U, 3U, 6U, and 12U.

Key Highlights: 1U = 10×10×10 cm • Mass: ~1.33 kg nominal • Standard Deployer: P-POD (Poly-Picosatellite Orbital Deployer)

What is Telemetry (TM)?

Derived from Greek 'tele' (remote) and 'metron' (measure). In spacecraft avionics, telemetry is the automated communication process by which onboard physical parameters (voltages, currents, temperatures, inertial motion, GPS coordinates, antenna deployments) are measured, digitized, and radioed down to Earth.

Key Highlights: Housekeeping Telemetry: Voltages, battery state of charge, subsystem thermal health • Payload Telemetry: Scientific, optical, or atmospheric experimental data • Downlink (TM) vs Uplink (TC - Telecommands)

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: What is the standard dimension and weight of a 1U CubeSat?

Ans: A standard 1U CubeSat measures 10 cm × 10 cm × 10 cm (100 mm cube) and has a nominal mass of up to 1.33 kg (often permitted up to 2 kg in modern standards).

Q2: What is the difference between Telemetry and Telecommand?

Ans: Telemetry (TM) is the downlink transmission of internal measurements and sensor data from satellite to ground. Telecommand (TC) is the uplink transmission of instructions/commands from ground to satellite.

1U STANDARD DIMENSIONS	1U STANDARD MASS	PRIMARY ORBIT	DOWNLINK FREQUENCY
100.0 ± 0.1 mm cube	**"d 1.33 kg (expanded to 2.0 kg)**	**LEO (Low Earth Orbit, 300–600 km)**	**VHF/UHF (435-438 MHz) / S-band (2.4 GHz)**

Uses of CubeSat

From High-Risk Commercial Missions to Interplanetary Exploration

CORE THESIS & DEFINITION:

CubeSats have transitioned from academic toys to frontline orbital workhorses, delivering low-cost Earth observation, space weather forecasting, IoT constellation relay, and deep-space interplanetary exploration.



1U CubeSat Standard Nanosatellite in Low Earth Orbit

1. Earth Observation & Remote Sensing

Constellations of 3U/6U CubeSats (like Planet Labs' Doves) capture daily multispectral imagery of the entire landmass of Earth. Applications include deforestation monitoring, agricultural NDVI crop yield tracking, wildfire detection, and coastal erosion analysis.

Key Highlights: Daily global revisit rates • Multispectral & hyperspectral bands • Disaster response imaging

2. Scientific Research & Space Weather

Upper atmosphere (thermosphere/ionosphere) sounding, solar flare radiation flux detection, plasma density profiling, and cosmic ray characterization using onboard miniaturized magnetometers and Geiger-Müller tubes.

Key Highlights: Ionospheric electron density • Solar geomagnetic storm monitoring • Microgravity crystal growth

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: Mention four major domains where CubeSats are actively used.

Ans: 1. Earth Observation & Agriculture 2. Space Weather & Ionosphere Science 3. Global IoT and Maritime/Aviation tracking (AIS/ADS-B) 4. In-orbit Technology Demonstration & COTS qualification.

Q2: What was the NASA MarCO mission?

Ans: Mars Cube One (MarCO) was the first interplanetary CubeSat mission (two 6U spacecraft) that flew past Mars in 2018, relaying telemetry from the InSight lander back to Earth during descent.

deep-space relay stations.

PLANET LABS DOVE CONSTELLATION	TYPICAL MISSION COST	ORBITAL LIFETIME	DEPLOYMENT METHOD
200+ 3U imaging satellites	**$50k - $500k (vs $500M+ traditional)**	**3 months to 5 years (LEO decay)**	**ISS Nanoracks / Rocket rideshares**

Basic Knowledge About Satellites

Orbital Mechanics, Gravity Balance & The 6 Core Subsystems

CORE THESIS & DEFINITION:

An artificial satellite is an object placed into orbit around a celestial body. It maintains its orbit through a continuous balance between gravitational attraction and the satellite's centrifugal inertia (perpetual free-fall).



Basic Knowledge About Satellites · Aerospace Technical Reference

How an Orbit Works: Newton's Cannonball

A satellite in circular orbit is continuously falling towards Earth due to gravity, but moves horizontally fast enough (~7.8 km/s in LEO) that the Earth curves away beneath it at the exact same rate.

Key Highlights: Perpetual free-fall condition • Gravity provides centripetal acceleration: Fg = Fc • LEO velocity ~28,000 km/h

The 6 Essential Subsystems of Every Satellite

1. OBC (On-Board Computer): Avionics brain / CD&H
2. EPS (Electrical Power System): Solar cells, LiFePO4 batteries, PMIC
3. ADCS (Attitude Determination & Control): Sensors & magnetorquers
4. COMM / TTC (Telemetry, Tracking & Command): Transceivers
5. Thermal & Structure: Chassis, Kapton insulation, radiating plates
6. Payload: The mission-specific instrument (camera, sensors, radar).

Key Highlights: OBC: Command & Data Handling • EPS: Power harvesting & regulation • ADCS: Pointing & Stabilization

MATHEMATICAL FORMULATIONS & PHYSICS DERIVATIONS

Circular Orbital Velocity

```
v = sqrt((G * M_E) / r)
```

Speed required to maintain a stable circular orbit at radius r from Earth's center.

Kepler's Third Law (Orbital Period)

```
T = 2 * pi * sqrt(r^3 / (G * M_E))
```

Calculates time to complete one full revolution around Earth.

EXAMINER VIVA QUESTIONS & DEFENSE ANSWERS

Q: Why doesn't a satellite fall straight down to Earth?

A: Because it has sufficient tangential velocity (~7.8 km/s in LEO). The gravitational force pulls it toward Earth, but Earth's surface curves away at the same rate, resulting in continuous circular free-fall.

CUBESAT 1U PROTOTYPE FLIGHT ROLE:

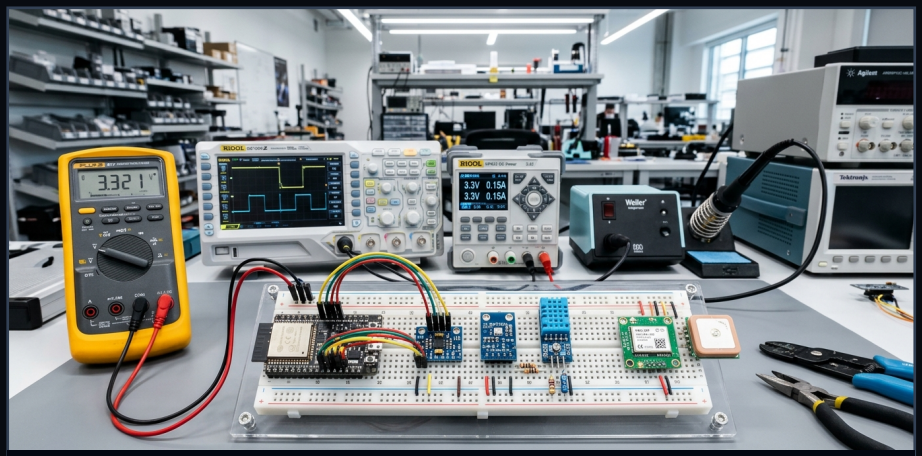
In our prototype, the ESP32 acts as the OBC subsystem, our breadboard rails emulate the EPS bus (regulated 3.3V), our sensors form the Payload and ADCS sensing suite, and the RF module provides the TTC subsystem.

Transmitter ESP: Onboard & Ground Station

Avionics Core, ESP-NOW Protocol & Uplink/Downlink Architecture

CORE THESIS & DEFINITION:

The onboard ESP32 acts as the flight transmitter and OBC. It samples sensors, packages telemetry packets, and transmits via 2.4 GHz RF (ESP-NOW/Wi-Fi). The Ground Station transmitter sends uplink telecommands.



Transmitter ESP: Onboard & Ground Station · Aerospace Technical Reference

Onboard ESP32 Transmitter Operations

1. Bus Master: Reads I2C sensors (MPU6050, BMP280) and UART sensors (NEO-6M GPS).
2. Packet Encoding: Compiles sensor variables into a compact binary struct or delimited telemetry frame with packet counter and CRC.
3. RF Modulation: Transmits via 2.4 GHz MAC-layer frames using Espressif's ESP-NOW protocol or LoRa transceiver module (SX1278).

Key Highlights: ESP-NOW: Peer-to-peer connectionless 2.4 GHz • Transmission latency < 2 ms • Zero router/ access-point dependency

Ground Station Transmitter (Uplink Role)

While the onboard ESP32 transmits telemetry down, the ground station ESP32 functions as a bidirectional transceiver. It transmits telecommands uplink: polling rate adjustments, sensor calibration triggers, emergency safe-mode beacon, or payload camera trigger.

Key Highlights: Uplink Command Authentication • Acknowledgement (ACK) handshake • Bi-directional telecommunication

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: What is ESP-NOW and why is it used in CubeSat prototyping?

Ans: ESP-NOW is a connectionless 2.4 GHz wireless communication protocol developed by Espressif. It allows peer-to-peer transmission using MAC addresses without needing a Wi-Fi router, offering low power, high speed (<2ms latency), and up to 250-byte payloads.

Q2: What does the onboard transmitter send to ground station?

Ans: It sends telemetry packets containing sensor readings (acceleration, angular velocity, pressure, altitude, temperature, humidity, GPS coordinates), system battery voltage, packet sequence ID, and CRC checksum.

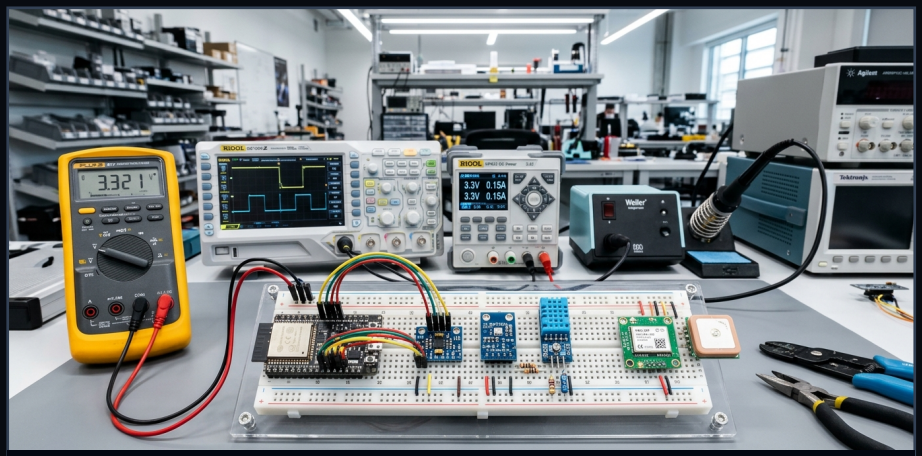
RF FREQUENCY	TX POWER	PAYLOAD SIZE (ESP-NOW)	LINE-OF-SIGHT RANGE
2.4 GHz ISM Band (Channels 1-13)	**+20 dBm (100 mW EIRP)**	**Up to 250 bytes per frame**	**200m–500m (directional patch > 1 km)**

Receiver ESP32 Type-C (Connected by USB)

The Physical Gateway Between Space Link & Ground Station PC

CORE THESIS & DEFINITION:

The Ground Station ESP32 Type-C module receives RF telemetry packets over the air, verifies packet integrity, and bridges the data directly to the ground laptop over high-speed USB-UART serial at 115200 baud.



Receiver ESP32 Type-C (Connected by USB) · Aerospace Technical Reference

RF Reception & Frame Decapsulation

Operates in promiscuous / receiver mode with its radio tuned to the CubeSat transmitter's channel and MAC address. Upon receiving an interrupt, the ESP32 receiver fires the onDataRecv callback in memory.

Key Highlights: Instant hardware interrupt trigger • RSSI (Signal Strength) measurement • Payload CRC verification

USB-to-UART Bridge Architecture

The onboard Type-C port is wired to a USB-to-UART bridge IC (such as CP2102, CH340C, or native USB CDC). It receives 5V power from the laptop's USB port, powers the ESP32 via a 3.3V LDO regulator, and forwards clean serial ASCII/JSON packets to the laptop.

Key Highlights: Baud rate: 115200 bps standard (up to 921600 bps) • USB CDC Virtual COM Port • Bidirectional pipeline for telemetry & commands

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: How does the ESP32 receiver communicate with the ground station laptop?

Ans: Via its USB Type-C connection. An on-board USB-to-UART bridge IC (such as CP2102 or CH340) converts the ESP32's hardware UART serial signals into USB CDC virtual COM port packets that the laptop reads.

Q2: What baud rate is typically used for the USB serial link?

Ans: 115200 baud (bits per second) is the industry standard for ESP32 telemetry, though speeds up to 921600 baud are supported for high-rate data.

CONNECTOR STANDARD	POWER SOURCE	LOGIC LEVEL	SERIAL PROTOCOL
USB 2.0 Type-C (Reversible)	**USB VBUS 5.0 V from Laptop**	**3.3V LVTTL (Transceiver to Bridge)**	**UART 8-N-1 (8 data bits, no parity, 1 stop bit)**

Laptop Display & Ground Station GUI

Real-Time Telemetry Visualization, Telecommand Console & Data Logging

CORE THESIS & DEFINITION:
The Ground Station Laptop acts as the flight controller's primary cockpit: parsing incoming serial packets, rendering 3D attitude vectors, plotting environmental telemetry strip-charts, and logging data to disk.

1. Real-Time Telemetry Parsing & Display

Software running on the laptop (Web Serial API, Python GUI with PyQt/Tkinter, or serial monitor) reads incoming JSON/CSV lines, validates framing, and maps values to real-time visual gauges: altitude altimeter, barometer dial, and temperature/humidity graphs.

Key Highlights: Sub-50ms UI update latency • Attitude Artificial Horizon indicator • Live GPS Ground-Track mapping

2. Telecommand Injection Console

Allows ground operators to type and send telecommands back across the USB link: adjust polling frequency, ping satellite for RTT (Round Trip Time) latency, calibrate sensor offsets, or trigger payload release.

Key Highlights: Uplink command verification • Real-time command history log • Audio/Visual threshold alarms

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: Why is a graphical display needed instead of just a raw serial terminal?

Ans: A graphical dashboard translates raw numerical telemetry into instant cognitive insights: artificial horizons show satellite tumble, gauge dials show pressure limits, and color-coded thresholds alert operators to anomalies before mission loss.

Q2: What is the role of data logging in satellite ground software?

Ans: Data logging saves every received telemetry frame with high-precision timestamps to CSV/database files for post-flight analysis, anomaly diagnosis, and orbital trajectory reconstruction.

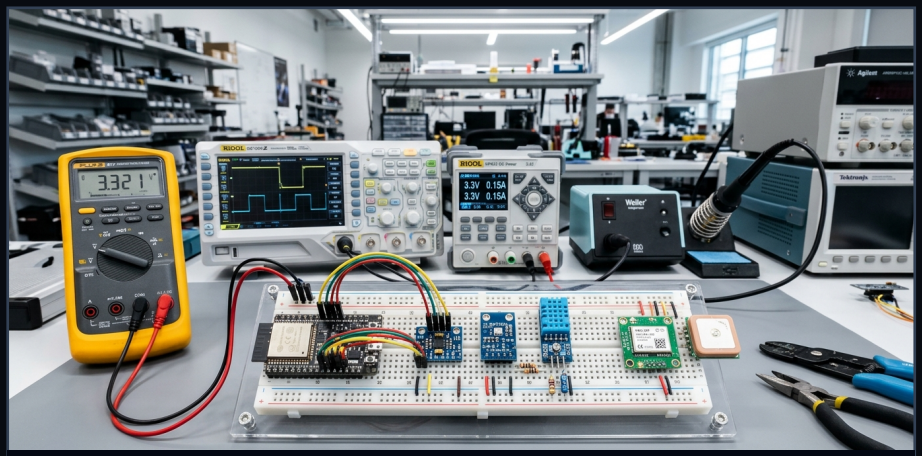
DATA INTERFACE	DISPLAY ELEMENTS	LOG FORMAT	ALERT TRIGGERS
Web Serial API / PySerial USB Bridge	**3D Horizon, Altimeter, GPS Map, Strip Charts**	**ISO-8601 Timestamped CSV & JSON Lines**	**High temp (>60°C), Battery critical (<3.2V)**

Sensors: MPU6050, BMP280, DHT11, NEO-6M & Antenna

Mathematical Formulas, Operating Principles & Prototype Implementation

CORE THESIS & DEFINITION:

The avionics sensor suite measures all primary flight variables: 6-DoF inertial motion (MPU6050), atmospheric pressure & altitude (BMP280), ambient humidity & temperature (DHT11), and GNSS navigation (NEO-6M + Ceramic Patch Antenna).



Sensors: MPU6050, BMP280, DHT11, NEO-6M & Antenna · Aerospace Technical Reference

1. MPU6050 (6-DoF IMU: Accelerometer + Gyroscope)

Combines a 3-axis MEMS accelerometer (detecting proof-mass deflection via differential capacitance) and a 3-axis MEMS gyroscope (detecting Coriolis force from vibrating ring). Communicates over I2C at address 0x68.

Key Highlights: Accelerometer: ±2g to ±16g • Gyroscope: ±250°/s to ±2000°/s • Onboard Digital Motion Processor (DMP)

2. BMP280 (High-Precision Barometer & Altimeter)

Piezoresistive silicon MEMS sensor measuring atmospheric pressure from 300 to 1100 hPa with ±0.12 hPa accuracy (~1 meter altitude resolution). I2C address 0x76 or 0x77.

Key Highlights: Altitude resolution: 1 m • RMS noise: 0.2 Pa • Ultra-low power: 2.7 µA @ 1 Hz

MATHEMATICAL FORMULATIONS & PHYSICS DERIVATIONS

BMP280 Hypsometric Barometric Altitude Formula

```
h = 44330 * (1 - (P / P0)^(1 / 5.255))
```

Calculates height above sea level based on the International Standard Atmosphere (ISA) pressure lapse rate.

MPU6050 Accelerometer Pitch & Roll Estimation

```
Roll (phi) = atan2(ay, az) * (180 / pi);  Pitch (theta) = atan2(-ax, sqrt(ay^2 + a
```

Derives tilt angles relative to Earth's 1g gravity vector from tri-axial acceleration measurements.

EXAMINER VIVA QUESTIONS & DEFENSE ANSWERS

Q: State the barometric altitude formula used by the BMP280 sensor.

A: h = 44330 * (1 - (P / 1013.25)^(1 / 5.255)), where P is the measured pressure in hPa and h is altitude in meters.

UART at 9600 baud.

CUBESAT 1U PROTOTYPE FLIGHT ROLE:

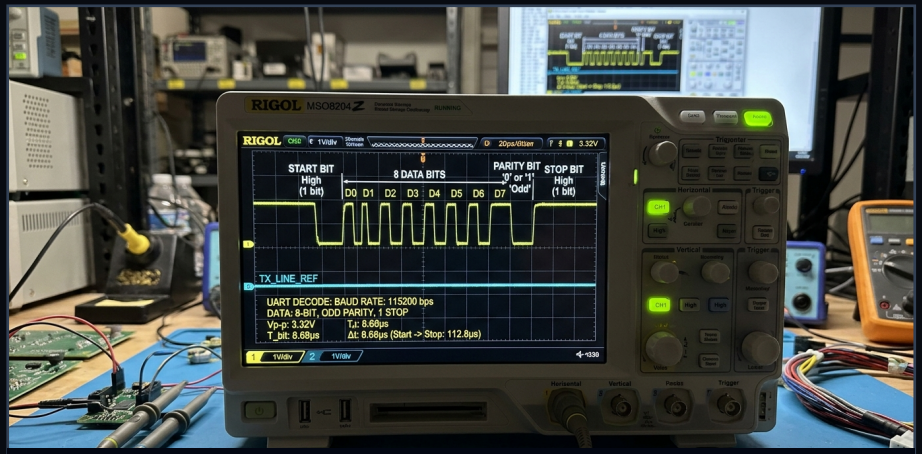
Provides the complete avionics payload: MPU6050 informs ADCS attitude control; BMP280 tracks ascent/altitude; DHT11 guards internal electronics compartment from moisture/overheating; NEO-6M provides geographic tracking.

What is a Protocol?

Rules of Engagement in Avionics Data Networks & Embedded Systems

CORE THESIS & DEFINITION:

A protocol is a standardized set of rules, formats, and signaling conventions that dictates how two or more electronic devices communicate, negotiate connections, package data, and handle errors.



Digital Storage Oscilloscope · UART Serial Signal Waveform & Timing Analysis

Core Functions of a Communication Protocol

1. Synchronization: How sender and receiver lock onto byte timings (clocked line vs agreed baud rate).
2. Addressing: Identifying which device on a shared bus is being spoken to (e.g. 7-bit I2C addresses).
3. Packet Framing: Preamble, header, payload length, and delimiter bytes.
4. Error Detection & Recovery: Parity bits, checksums, and CRCs to detect corrupted bits.

Key Highlights: Syntax: Data format & coding • Semantics: Meaning of command bits • Timing: Clock synchronization & latency

Protocol Comparison in CubeSat Avionics

• I2C (Inter-Integrated Circuit): Synchronous, 2-wire (SDA/SCL), multi-master/multi-slave, up to 400 kHz. Used for BMP280, MPU6050.
• SPI (Serial Peripheral Interface): Synchronous, 4-wire (MOSI/MISO/SCK/CS), high-speed (up to 50+ MHz). Used for SD cards & LoRa radios.
• UART (Universal Asynchronous Receiver-Transmitter): Asynchronous, 2-wire (TX/RX), point-to-point. Used for GPS & USB console.
• Wireless Protocols: ESP-NOW, LoRa, AX.25 packet radio.

Key Highlights: I2C: Low wire count, shared sensor bus • SPI: Maximum bandwidth • UART: Simple point-to-point streaming

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: Define a communication protocol in embedded systems.

Ans: A communication protocol is a predefined set of rules, data formats, and timing standards that allow disparate electronic devices to reliably transmit, receive, interpret, and verify data across a physical communication medium.

Q2: Compare I2C and UART protocols in two key aspects.

Ans: 1. Clocking: I2C is synchronous (uses a dedicated SCL clock wire), whereas UART is asynchronous (has no clock wire and relies on matching baud rates). 2. Topology: I2C supports multiple addressable devices on just two wires, whereas UART is strictly point-to-point between two devices.

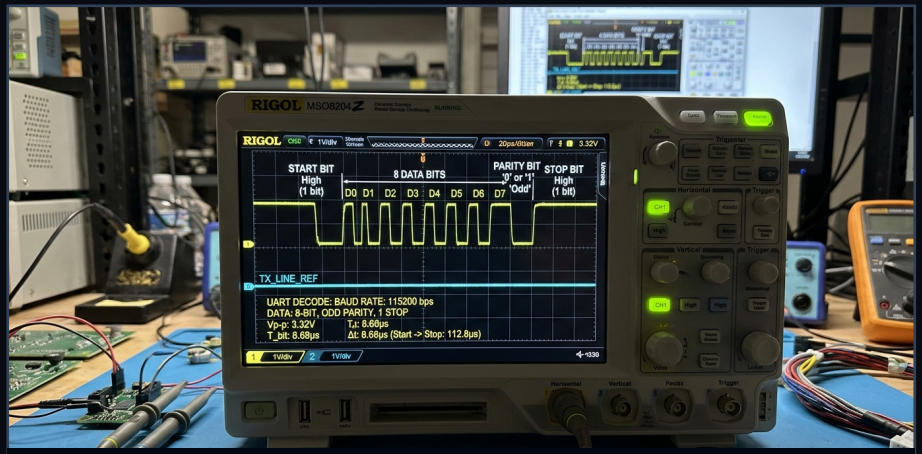
I2C SPEED	SPI SPEED	UART SPEED	ERROR DETECTION
Standard: 100 kHz, Fast Mode: 400 kHz	**10 MHz - 80 MHz**	**9600 to 921600 baud**	**Parity, Checksum, CRC-16, CRC-32**

UART Basics (Universal Asynchronous Receiver-Transmitter)

Bit Timing, Packet Framing Structure & Hardware Implementation

CORE THESIS & DEFINITION:

UART is a fundamental hardware serial protocol that converts parallel byte data from a microcontroller CPU into sequential serial bits for transmission over a single pair of TX/RX wires with no shared clock line.



Digital Storage Oscilloscope · UART Serial Signal Waveform & Timing Analysis

Why 'Asynchronous' Matters

Because there is no separate clock wire between transmitter and receiver, both ends must be configured in advance to the exact same Baud Rate (bits per second). Each bit has an exact time duration: T_bit = 1 / BaudRate.

Key Highlights: No shared clock line required • Cross-connected wires: TX -> RX, RX -> TX, Common GND • Common bauds: 9600 (GPS), 115200 (ESP32 USB console)

The Anatomy of a UART Packet Frame

1. Idle Line: Voltage stays at logic HIGH (3.3V / Mark state).
2. Start Bit: Driven LOW (0V / Space state) for exactly 1 bit period. This falling edge wakes up the receiver's internal timer.
3. Data Bits: 5 to 8 bits transmitted sequentially, starting with the Least Significant Bit (LSB first).
4. Parity Bit (Optional): Even/Odd parity bit to detect single-bit corruption.
5. Stop Bit(s): Driven logic HIGH for 1 or 2 bit periods to allow receiver framing recovery.

Key Highlights: Total Frame: 1 Start + 8 Data + 0 Parity + 1 Stop = 10 bits per byte • Baud 115200 => 11,520 bytes/sec Maximum throughput • T_bit @ 115200 = 8688 microseconds

MATHEMATICAL FORMULATIONS & PHYSICS DERIVATIONS

UART Bit Duration & Effective Transfer Rate

```
T_bit = 1 / Baud_Rate;  Byte_Rate = Baud_Rate / (Start + Data + Parity + Stop)
```

Calculates the electrical pulse duration of a single bit and total usable byte throughput.

EXAMINER VIVA QUESTIONS & DEFENSE ANSWERS

Q: Explain the structure of a standard 8-N-1 UART frame.

A: A standard 8-N-1 UART frame consists of 10 bits: 1 Start bit (logic LOW 0V), 8 Data bits (transmitted LSB to MSB), No parity bit, and 1 Stop bit (logic HIGH 3.3V).

CUBESAT 1U PROTOTYPE FLIGHT ROLE:

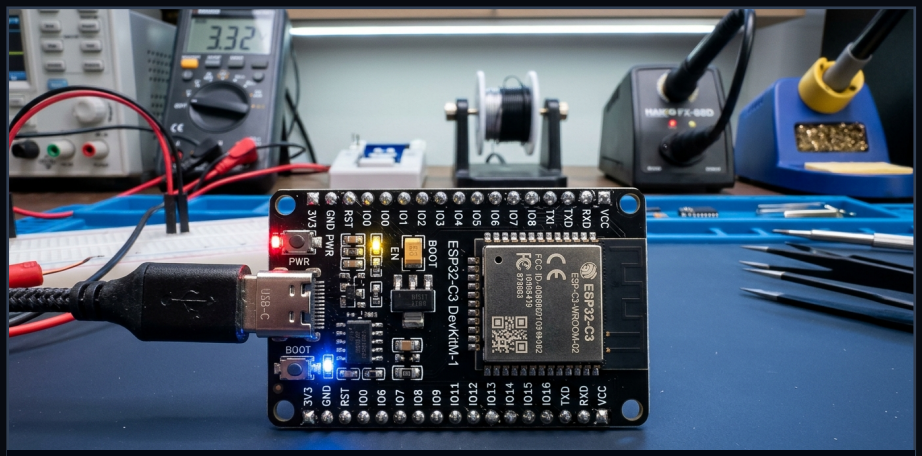
UART connects the NEO-6M GPS receiver to the ESP32 (UART2 on GPIO16/17), and bridges the Ground Station ESP32 to the Laptop serial console (UART0 on GPIO1/3 via Type-C bridge).

About Microcontrollers (MCU)

Architecture, SoC Integration & Role as Satellite On-Board Computer

CORE THESIS & DEFINITION:

A microcontroller (MCU) is a self-contained System-on-Chip (SoC) combining processor cores, volatile RAM, non-volatile Flash, and configurable I/O peripherals onto a single silicon substrate, tailored for real-time control.



ESP32 Dual-Core Type-C DevKit · Hardware & GPIO Avionics Bus

Microcontroller (MCU) vs Microprocessor (MPU)

• Microprocessor (e.g. Intel Core i7, Raspberry Pi BCM2711): High computational throughput, requires external RAM and Flash, runs complex OS (Linux/Windows), high power (watts).
• Microcontroller (e.g. ESP32, STM32, ATmega328): Integrated RAM and Flash, deterministic sub-microsecond interrupt latency, ultra-low power (milliwatts), executes code directly on bare metal or real-time OS (FreeRTOS).

Key Highlights: Integrated on-chip peripherals • Deterministic real-time execution • Low power consumption

Essential Peripherals Inside an MCU

1. GPIO (General Purpose Input/Output): Digital read/write pins.
2. Hardware Timers & PWM: Microsecond task scheduling and motor control.
3. ADC / DAC: Analog-to-digital and digital-to-analog conversion.
4. Serial Controllers: Dedicated silicon silicon hardware for UART, I2C, SPI.
5. Watchdog Timer (WDT): Hardware failsafe that resets the MCU if code freezes.

Key Highlights: Watchdog Timer: Autonomous reboot upon lockup • DMA (Direct Memory Access) • Hardware interrupts

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: What is the difference between a microcontroller and a microprocessor?

Ans: A microcontroller integrates the CPU, RAM, ROM/Flash, and I/O peripherals on a single silicon chip and is optimized for low-power, real-time control. A microprocessor contains only the CPU core and requires external RAM, ROM, and peripheral chips.

Q2: What is a Watchdog Timer (WDT) and why is it critical in a CubeSat?

Ans: A Watchdog Timer is a hardware countdown timer that automatically resets the microcontroller if the software fails to refresh ('kick') it in time. In CubeSats, where radiation can cause code execution to deadlock, the WDT guarantees autonomous recovery.

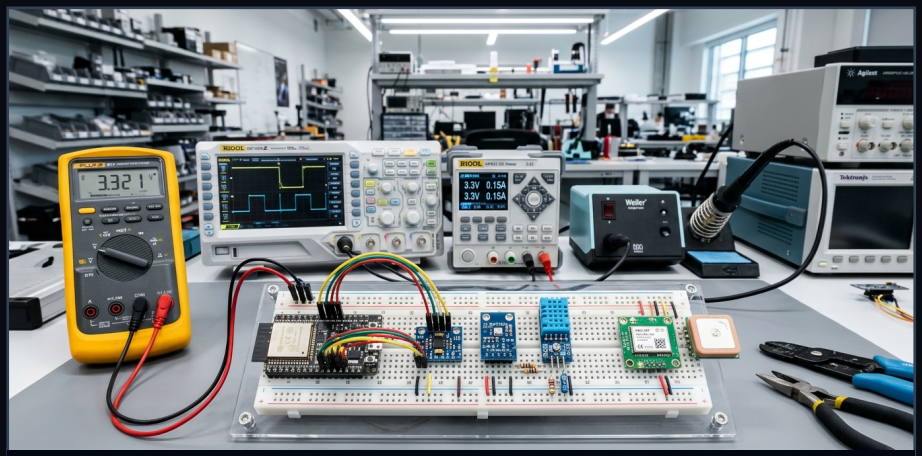
CORE TYPES	POWER MODES	CLOCK FREQUENCIES	TYPICAL SPACE FLAVOUR
Harvard Architecture, RISC (ARM, Xtensa, RISC-V)	Active, Light Sleep, Deep Sleep (µA), Hibernation	16 MHz (ATmega) to 240 MHz (ESP32)	Rad-tolerant STM32 / Microchip SAMV71 / COTS ESP32

Important Page on ESP32 Type-C DevKit

Silicon Specifications, Dual-Core Xtensa LX6 & Pinout Architecture

CORE THESIS & DEFINITION:

The ESP32 Type-C Development Board is the primary computing and communications engine of our CubeSat. It features a dual-core 32-bit processor at 240 MHz, integrated 2.4 GHz Wi-Fi/ Bluetooth, and a robust USB Type-C interface.



Important Page on ESP32 Type-C DevKit · Aerospace Technical Reference

Processor Architecture & Memory

- CPU: Tensilica Xtensa Dual-Core 32-bit LX6 microprocessor operating up to 240 MHz (~600 DMIPS computing power).
- Core 0: Assigned to RF, Wi-Fi, and ESP-NOW communication stack.
- Core 1: Assigned to application code, sensor polling, and math filters.
- Memory: 520 KB internal SRAM, 448 KB ROM, and 4 MB to 16 MB external QSPI Flash memory.

Key Highlights: Dual-core enables non-blocking sensor acquisition • Built-in FreeRTOS preemptive kernel • ULP (Ultra-Low-Power) co-processor

Type-C Hardware Interface & Power Subsystem

- USB Type-C: Reversible connector providing 5V DC and USB 2.0 Full Speed data.
- LDO Voltage Regulator: Down-converts 5V USB to clean 3.3V logic (typically AMS1117 or ME6211 rated up to 800 mA to handle Wi-Fi peak TX bursts).
- USB-to-UART Bridge: CH340 or CP2102 IC converting PC serial to GPIO1 (TX0) and GPIO3 (RX0).
- Boot & Reset: Integrated auto-programming circuit using DTR/RTS lines to pulse EN (Reset) and IO0 into flashing mode.

Key Highlights: 5V to 3.3V LDO regulator on board • Auto-reset / flashing circuit • 34 programmable GPIOs

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: What is the CPU architecture and maximum clock speed of the ESP32?

Ans: The ESP32 is powered by a dual-core Tensilica Xtensa 32-bit LX6 microprocessor operating at up to 240 MHz.

Q2: What are the advantages of having a dual-core CPU in an avionics OBC?

Ans: Core 0 can run background RF communications (ESP-NOW/Wi-Fi) while Core 1 executes time-critical sensor sampling, attitude estimation, and flight logic without interruption or latency spikes.

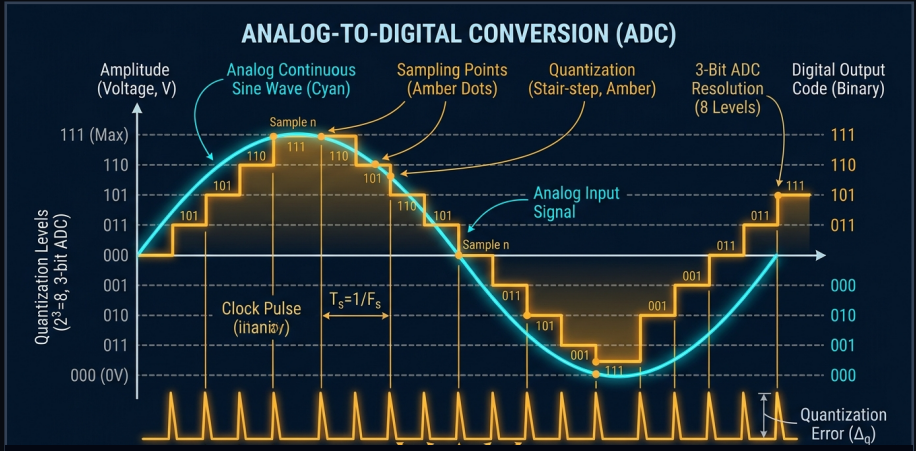
PROCESSOR CORE	INTERNAL SRAM	FLASH MEMORY	ADC RESOLUTION
Xtensa 32-bit LX6 Dual-Core @ 240 MHz	**520 KB SRAM**	**4 MB (up to 16 MB) SPI Flash**	**12-bit (up to 18 channels, 0-4095)**

About ADC (Analog-to-Digital Converter)

Quantization Theory, LSB Resolution & ESP32 12-Bit Architecture

CORE THESIS & DEFINITION:

An ADC converts continuous, real-world analog voltage levels (such as from a battery monitor, solar cell, or thermistor) into discrete binary numerical values that a digital microcontroller can process.



Analog-to-Digital Conversion (ADC) · Signal Sampling & 12-bit Quantization Stair-Step

Sampling & Quantization Principles

1. Sampling: Measuring the analog voltage at discrete time intervals.
2. Quantization: Mapping the infinite analog voltage domain into a finite number of digital discrete levels determined by ADC resolution (n bits): Total Levels = 2^n.
3. LSB (Least Significant Bit): The smallest voltage change detectable by the ADC: LSB = Vref / (2^n).

Key Highlights: 12-bit ADC = 2^12 = 4096 discrete steps (0 to 4095) • LSB resolution = 3.3V / 4096 "H 0.805 mV / count • Quantization error = ± 1/2 LSB

ESP32 ADC Features & Attenuation Modes

The ESP32 features two 12-bit SAR (Successive Approximation Register) ADCs: ADC1 (8 channels on GPIO32-39) and ADC2 (10 channels, shared with Wi-Fi). To measure up to 3.3V, an input attenuator is configured:

- 0 dB: 0 to 1.1 V (full scale)
- 2.5 dB: 0 to 1.5 V
- 6 dB: 0 to 2.2 V
- 11 dB: 0 to 3.3 V (recommended for 3.3V rail & battery divider monitoring).

Key Highlights: ADC1 safe during Wi-Fi; ADC2 conflicts with Wi-Fi • 11 dB attenuation extends range to 3.3V • Internal eFuse voltage calibration

MATHEMATICAL FORMULATIONS & PHYSICS DERIVATIONS

ADC Quantization Formula

```
Digital_Count = floor((V_in / V_ref) * (2^n - 1))
```

Converts an input voltage into a raw digital integer count.

Reconstructed Voltage Formula

```
V_in = (Digital_Count / 4095) * 3.3 V
```

Calculates the physical voltage measured by the ADC pin.

EXAMINER VIVA QUESTIONS & DEFENSE ANSWERS

Q: What is the resolution and range of the ESP32's ADC?

A: The ESP32 features a 12-bit ADC resolution, providing 4096 discrete levels (counts from 0 to 4095), giving a voltage step (LSB) of approximately 0.805 mV when referenced to 3.3V.

CUBESAT 1U PROTOTYPE FLIGHT ROLE:

Monitors CubeSat battery voltage (LiPo 3.7V - 4.2V stepped down via a 2:1 resistive divider to 0-2.1V), solar panel current sensors, and NTC thermistors.

Blynk IoT Cloud Platform

Cloud Telemetry Dashboards, Mobile HMI & Global Data Routing

CORE THESIS & DEFINITION:

Blynk IoT is a low-latency IoT cloud ecosystem that bridges the ESP32 Ground Station to cloud dashboards and mobile smartphone apps, enabling global remote tracking of CubeSat telemetry with zero server overhead.



Blynk IoT Cloud Platform · Mobile Telemetry Dashboard & Live Gauges

Blynk IoT Architecture

1. Device Hardware: ESP32 running the Blynk library connects to internet via Wi-Fi.
2. Blynk Cloud Server: Secure cloud broker managing device state, datastreams, and time-series telemetry storage.
3. User Interface: Customizable web console and iOS/Android mobile dashboard with gauges, interactive charts, GPS maps, and push notifications.

Key Highlights: Cloud-based remote access • Multi-device real-time sync • Automated email/push anomaly alerts

The Concept of 'Virtual Pins' (V-Pins)

Virtual Pins (V0 to V255) are software channels that decouple physical hardware pins from dashboard widgets. Instead of wiring a widget directly to GPIO4, the code writes telemetry variables to Virtual Pins:

- V0: CubeSat Altitude (m)
- V1: Barometric Pressure (hPa)
- V2: Battery Voltage (V)
- V3: Temperature (°C)
- V4: GPS Location (Lat/Lon map pin)
- V5: Pitch / Roll attitude

Key Highlights: Blynk.virtualWrite(V0, altitude) • Hardware abstraction layer • Bi-directional widget sliders for telecommands

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: What are Virtual Pins in Blynk IoT?

Ans: Virtual Pins (V0-V255) are software abstraction channels used to exchange variables, sensor telemetry, and control commands between the microcontroller firmware and Blynk cloud dashboard widgets without binding to physical GPIO pins.

Q2: How does Blynk IoT enhance a CubeSat mission?

Ans: It allows the ground station to push received telemetry to the cloud, giving ground operators, students, and research teams real-time access to flight dashboards on web browsers and smartphones from anywhere globally.

CLOUD PROTOCOL	VIRTUAL PIN RANGE	SYNC LATENCY	SUPPORTED PLATFORMS
Lightweight Binary TCP / HTTPS / MQTT	**V0 to V255 (256 independent channels)**	**< 100 ms globally**	**Web Browser, iOS, Android Apps**

IoT Sensors Overview & Classification

Transducer Fundamentals, Environmental Sensing & Space Environmental Demands

CORE THESIS & DEFINITION:

IoT sensors are electronic transducers that convert physical, mechanical, thermal, or chemical phenomena into readable electrical signals (voltage, current, or digital packets) for embedded microcontrollers.



Blynk IoT Cloud Platform · Mobile Telemetry Dashboard & Live Gauges

Transducer Classification: Analog vs Digital

• Analog Sensors: Output a continuous varying voltage proportional to the measured property (e.g. NTC thermistors, LDR light sensors, analog strain gauges). Require an on-chip ADC to digitize.
• Digital Sensors: Incorporate an onboard sensing element, analog front-end, ADC, and digital signal processor (DSP). They communicate measured values directly over digital buses like I2C, SPI, or UART (e.g. BMP280, MPU6050, NEO-6M).

Key Highlights: Analog: Simple, passive, noise-susceptible • Digital: Calibrated, integrated DSP, noise-immune bus

Crucial Performance Metrics for Space Sensors

1. Sensitivity: Ratio of electrical output change to physical input change.
2. Accuracy vs Precision: Accuracy is proximity to true value; precision is repeatability.
3. Operating Temperature: Space components face extreme thermal swings (-40°C to +85°C in LEO).
4. Outgassing & Radiation Tolerance: Must not release volatile chemicals in hard vacuum or suffer single-event upsets.

Key Highlights: Thermal cycling resistance • Low outgassing in vacuum • Radiation latch-up immunity

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: Differentiate between an analog sensor and a digital sensor.

Ans: An analog sensor produces a continuous voltage or current that must be converted by an ADC, while a digital sensor has an internal ADC and DSP that converts physical data into calibrated digital packets over I2C, SPI, or UART.

Q2: Why is operating temperature range so critical for satellite sensors?

Ans: In Low Earth Orbit, a satellite passes between direct unattenuated solar radiation and Earth's cold eclipse shadow every 90 minutes, exposing components to extreme temperature fluctuations from -40°C to +85°C.

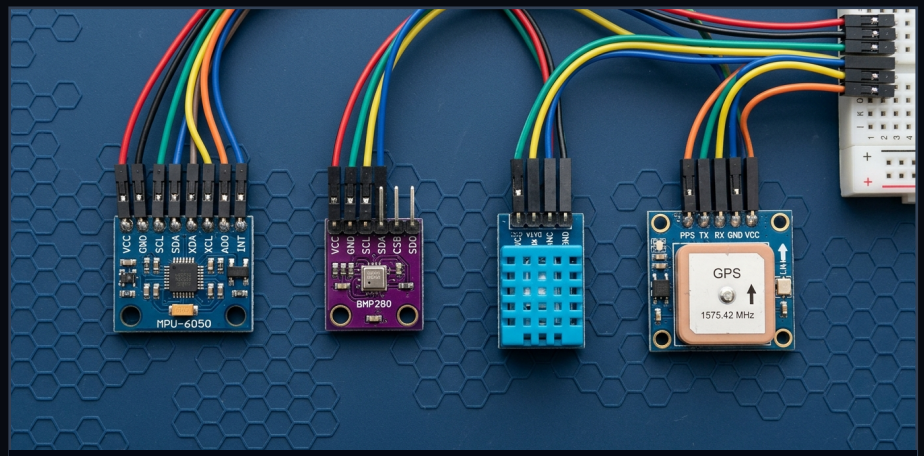
ANALOG SENSOR OUTPUT	DIGITAL SENSOR BUS	THERMAL RANGE (INDUSTRIAL)	SPACE VACUUM THRESHOLD
0 to 3.3V continuous voltage	**I2C (SDA/SCL) or SPI (MOSI/MISO)**	**-40°C to +85°C**	**< 10^-5 Torr**

Thermistor & Thermal Sensing

NTC Physics, Voltage Divider Circuits & The Steinhart-Hart Equation

CORE THESIS & DEFINITION:

A thermistor (thermally sensitive resistor) is a semiconductor resistor whose electrical resistance exhibits large, predictable changes in response to temperature variations. Essential for monitoring satellite battery and solar array thermal health.



Avionics Sensor Suite: MPU-6050 (6-DoF IMU), BMP280, DHT11 & GPS Patch Array

NTC vs PTC Thermistors

• NTC (Negative Temperature Coefficient): Electrical resistance DECREASES exponentially as temperature increases. Widely used for precision temperature measurement in CubeSat electronics bays.
• PTC (Positive Temperature Coefficient): Resistance INCREASES as temperature rises. Commonly used as self-resetting over-current fuses (polyswitches) in power distribution boards.

Key Highlights: NTC: R decreases with rising T • High sensitivity (typically -4% / °C) • Small thermal mass for fast response

The Voltage Divider Readout Circuit

Because microcontrollers cannot measure resistance directly, the NTC thermistor is placed in a series voltage divider with a known precision reference resistor R0 (typically 10 k:'Â R FöÆW ance). The divided voltage V_out is sampled by the ESP32 ADC.

*Key Highlights: V_out = Vcc * (R0 / (R_ntc + R0)) • R_ntc = R0 * (Vcc / V_out - 1) • Linearization via mathematical modeling*

MATHEMATICAL FORMULATIONS & PHYSICS DERIVATIONS

Beta (Beta) Parameter Temperature Equation

```
1 / T = 1 / T0 + (1 / beta) * ln(R / R0)
```

Standard equation used to calculate absolute temperature in Kelvin from measured NTC resistance.

Steinhart-Hart Equation (High-Precision Model)

```
1 / T = A + B * ln(R) + C * (ln(R))^3
```

Third-order polynomial model providing sub-0.05°C accuracy across wide temperature bands.

EXAMINER VIVA QUESTIONS & DEFENSE ANSWERS

Q: What does NTC stand for, and how does its resistance behave?

A: NTC stands for Negative Temperature Coefficient. Its electrical resistance decreases exponentially as temperature increases.

CUBESAT 1U PROTOTYPE FLIGHT ROLE:

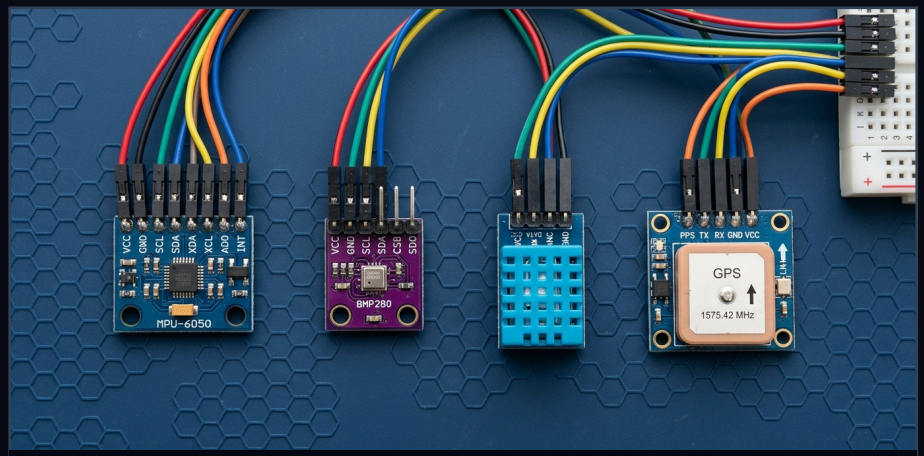
Mounted directly to the Li-ion battery pack (charging must halt below 0°C or above 45°C to prevent thermal runaway), solar panel backside, and radio power amplifiers.

Types of Motion & Gyroscope Physics

6 Degrees of Freedom, The Coriolis Effect & Attitude Determination

CORE THESIS & DEFINITION:

Spacecraft move in 6 Degrees of Freedom (6-DoF): 3 translational motions along coordinate axes, and 3 rotational motions around those axes. The gyroscope measures rotational angular rates using the Coriolis effect.



Avionics Sensor Suite: MPU-6050 (6-DoF IMU), BMP280, DHT11 & GPS Patch Array

6 Degrees of Freedom (6-DoF) in Spaceflight

- Translational Motions (Linear, m/s):
  1. Surge (X-axis: forward / backward along velocity vector)
  2. Sway (Y-axis: lateral left / right drift)
  3. Heave (Z-axis: vertical up / down orbital altitude change)
- Rotational Motions (Angular, deg/s):
  1. Roll (phi, rotation around longitudinal X-axis)
  2. Pitch (theta, rotation around lateral Y-axis)

Key Highlights: Roll, Pitch, Yaw dictate antenna and solar pointing • Detumbling: Cancelling post-separation angular momentum

How a MEMS Gyroscope Works: The Coriolis Force

Inside a MEMS gyroscope, a tiny silicon proof-mass is kept in continuous high-frequency vibration along an internal drive axis. When the CubeSat rotates, the Coriolis acceleration deflects the vibrating mass along a perpendicular sense axis, measured by capacitive finger combs.

Key Highlights: No rotating spinning wheels required • Vibrating ring silicon MEMS • Output: Angular velocity omega (deg/s)

MATHEMATICAL FORMULATIONS & PHYSICS DERIVATIONS

Coriolis Acceleration Vector

```
a_coriolis = 2 * (v_drive x Omega)
```

The apparent inertial acceleration experienced by a vibrating mass in a rotating frame of reference.

Angular Orientation from Gyro Integration

```
theta(t) = theta(t-1) + omega * Delta_t
```

Discrete time numerical integration of angular velocity to track rotation angle.

EXAMINER VIVA QUESTIONS & DEFENSE ANSWERS

Q: Define the 6 Degrees of Freedom (6-DoF) of a satellite.

A: 6-DoF includes 3 translational motions along the X, Y, and Z axes (Surge, Sway, Heave) and 3 rotational motions around those axes (Roll, Pitch, and Yaw).

CUBESAT 1U PROTOTYPE FLIGHT ROLE:

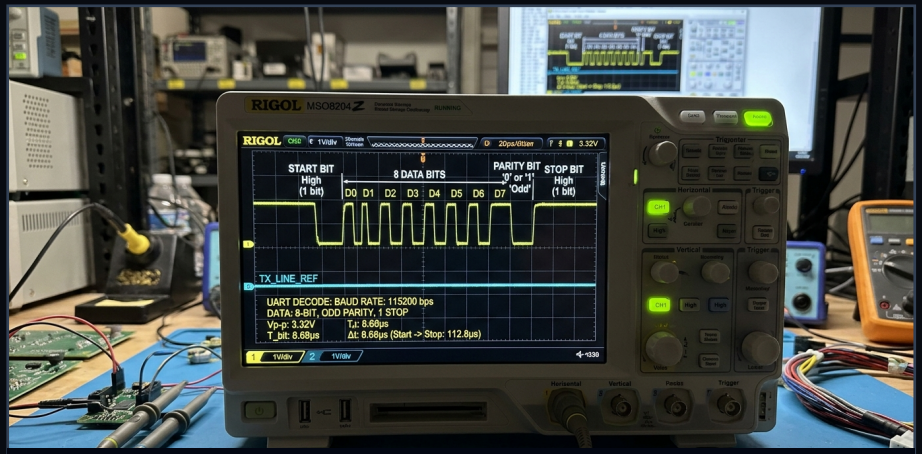
Critical for ADCS (Attitude Determination and Control System). Immediately after separation from the launch vehicle P-POD deployer, the satellite 'tumbles' wildly; the gyroscope detects spin rates to trigger magnetic detumbling coils (magnetorquers).

System Block Diagram

End-to-End Avionics, Communications & Ground Station Architecture

CORE THESIS & DEFINITION:

The complete system architecture diagram illustrates how power, sensor buses, microcontroller processing, RF transmission, USB serial bridging, and cloud monitoring interconnect seamlessly.



Digital Storage Oscilloscope · UART Serial Signal Waveform & Timing Analysis

Space Segment: On-Board Avionics

- Power: 3.7V Li-ion battery regulated to 3.3V.
- Sensors: MPU6050 (I2C 0x68), BMP280 (I2C 0x76), DHT11 (GPIO4 1-Wire), NEO-6M GPS (UART2 GPIO16/17), NTC Thermistor (ADC1 GPIO34).
- On-Board Computer: ESP32 Transmitter compiling data into binary telemetry frames and modulating over 2.4 GHz ESP-NOW.

Key Highlights: I2C Sensor Bus (SDA GPIO21, SCL GPIO22) • UART2 GPS Link • ESP-NOW 2.4 GHz RF Link

Ground Segment: Receiver & Displays

- Receiver Gateway: ESP32 Type-C receiver capturing RF frames via interrupt callback.
- Physical Link: USB Type-C cable with CP2102/CH340 USB-UART bridge at 115200 baud.
- Mission Control: Ground Station Laptop running GUI dashboard, serial parser, and CSV data logger.
- Cloud IoT: Internet gateway forwarding packets to Blynk Cloud for mobile monitoring.

Key Highlights: USB CDC Serial COM link • Laptop Dashboard GUI • Blynk IoT Cloud Sync

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: Explain the data flow from physical satellite sensor to the ground operator's screen.

Ans: 1. Sensor converts physical parameter to electrical signal -> 2. Transmitted over I2C/UART/ADC to onboard ESP32 OBC -> 3. Packed into telemetry frame and transmitted via 2.4 GHz RF -> 4. Received by ground ESP32 -> 5. Forwarded via USB Type-C serial to laptop -> 6. Parsed and rendered on dashboard GUI and pushed to Blynk Cloud.

Q2: What buses and protocols are used in this CubeSat architecture?

Ans: I2C for MPU6050 and BMP280; UART for NEO-6M GPS and USB PC serial; 1-Wire for DHT11; ADC for Thermistor; ESP-NOW for RF wireless telemetry; and TCP/IP for Blynk Cloud.

INTERNAL BUSES	WIRELESS LINK	HOST BRIDGE	CLOUD SYNC
I2C (Fast 400kHz), UART (9600 & 115200)	**2.4 GHz ISM Band (ESP-NOW)**	**USB Type-C (Virtual COM Port)**	**Wi-Fi HTTPS/MQTT to Blynk IoT**

Ground Station & Display Architecture

Antenna Front-End, Software Stack, Link Budgets & Real-Time Telemetry

CORE THESIS & DEFINITION:

The Ground Station is the terrestrial nerve center of the space mission. It provides RF signal reception, packet decapsulation, coordinate tracking, real-time telemetry rendering, and uplink commanding.



Ground Station & Display Architecture · Aerospace Technical Reference

Ground Station Architecture Layers

1. RF Front-End & Antenna: High-gain directional Yagi or patch antenna with low-noise amplifier (LNA).
2. Hardware Receiver: ESP32 Type-C node executing packet validation and RSSI signal measurement.
3. Serial Demux & Packet Decapsulation: Stripping start/stop delimiters, verifying CRC checksums.
4. Visual Telemetry HUD: Real-time gauges, 3D attitude indicator, strip charts, and audible alarms.
5. Command Uplink & Storage: Database logging and telecommand dispatching.

Key Highlights: Antenna gain & tracking rotor • Hardware packet validation • Comprehensive flight visualization

Link Budget & Signal Strength (RSSI)

To ensure packets can be received across the space-to-ground link, the Received Power (P_rx) must exceed receiver sensitivity (typically -97 dBm for ESP32):
P_rx = P_tx + G_tx + G_rx - FSPL, where FSPL is Free Space Path Loss: FSPL(dB) = 20*log10(d) + 20*log10(f) + 92.45.

Key Highlights: RSSI: Received Signal Strength Indicator in dBm • Packet Error Rate (PER) tracking • Link margin safety buffer

MATHEMATICAL FORMULATIONS & PHYSICS DERIVATIONS

Free Space Path Loss (FSPL) Formula

```
FSPL (dB) = 20 * log10(d_km) + 20 * log10(f_MHz) + 32.44
```

Calculates the radio signal power attenuation propagating through free space over distance d.

EXAMINER VIVA QUESTIONS & DEFENSE ANSWERS

Q: What is a Ground Station and what are its primary responsibilities?

A: A Ground Station is the terrestrial tracking and communications facility that receives telemetry downlinks from the satellite, tracks its orbital path, verifies data integrity, displays system health to flight controllers, logs data for science, and transmits telecommands uplink.

CUBESAT 1U PROTOTYPE FLIGHT ROLE:

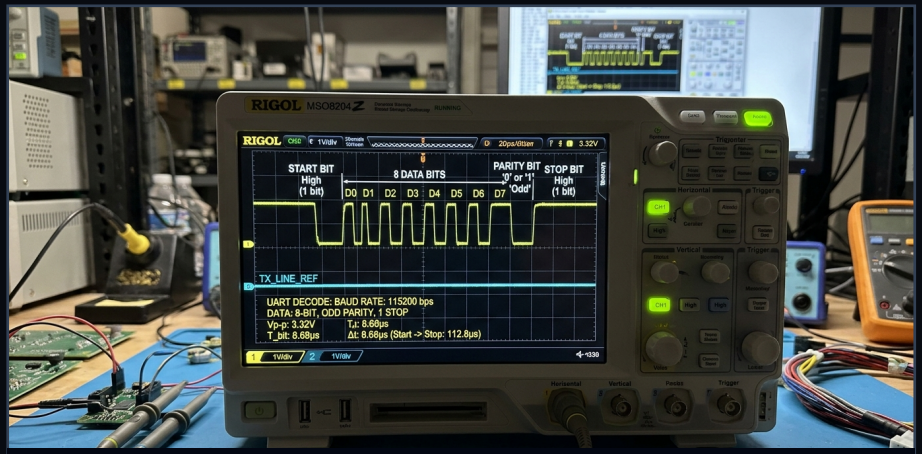
Forms the operational base station used by students, engineers, and ground controllers to monitor flight readiness, payload success, and system health.

MAC Address vs IP Address

Layer 2 Hardware Identifiers vs Layer 3 Network Routing in CubeSat Avionics

CORE THESIS & DEFINITION:

A MAC address is a 48-bit physical hardware identifier burned into silicon for local device-to-device communication (Layer 2). An IP address is a 32-bit (IPv4) or 128-bit (IPv6) logical address assigned to route data across networks and the Internet (Layer 3).



Digital Storage Oscilloscope · UART Serial Signal Waveform & Timing Analysis

1. MAC Address (Media Access Control) in ESP32

- Format: 48-bit binary value expressed as 6 hexadecimal octets (e.g. 24:6F:28:B4:7A:10).
- Structure: First 24 bits are the Organizationally Unique Identifier (OUI) assigned to Espressif; last 24 bits are the unique device serial.
- Invariance: Burned permanently into the ESP32's internal eFuse memory during chip fabrication.

Key Highlights: OSI Layer 2 (Data Link Layer) • 48-bit globally unique identifier • Hardware-burned into silicon eFuse

2. Why ESP-NOW Uses MAC Addresses (No IP Overhead)

ESP-NOW operates directly on Layer 2 using raw IEEE 802.11 vendor-specific action frames addressed to the receiver's MAC address. It completely bypasses IP routing, DHCP negotiation, ARP tables, and subnet masks. This enables sub-2ms packet delivery and minimal power consumption for satellite downlinks.

Key Highlights: Zero IP/TCP/UDP protocol overhead • Transmission latency < 2 ms • Direct peer-to-peer radio binding

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: What is a MAC address and how many bits does it consist of?

Ans: A MAC (Media Access Control) address is a 48-bit (6-byte) physical hardware identifier burned into the network interface card (or ESP32 eFuse) that uniquely identifies a physical device on the Data Link layer (OSI Layer 2).

Q2: Why does ESP-NOW use MAC addresses instead of IP addresses for CubeSat telemetry?

Ans: Because ESP-NOW is a connectionless Layer-2 protocol. Bypassing IP routing, DHCP discovery, and ARP resolution eliminates protocol overhead, enabling instant transmission with latency under 2 ms and significantly lower battery power consumption.

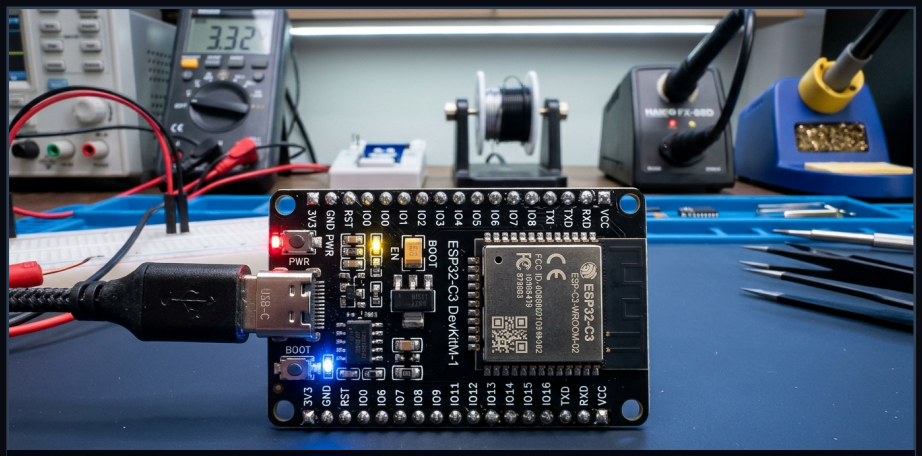
MAC ADDRESS LENGTH	IPV4 ADDRESS LENGTH	OSI ARCHITECTURE	AVIONICS ROLE
48 bits (6 hex octets)	**32 bits (4 decimal octets)**	**MAC: Layer 2 | IP: Layer 3**	**MAC: ESP-NOW RF Link | IP: Cloud IoT / Open MCT**

Ceramic Patch Antenna & Spacecraft RF Antennas

High-Dielectric Miniaturization, RHCP Polarization & Resonant Length Physics

CORE THESIS & DEFINITION:

A Ceramic Patch Antenna utilizes a high-permittivity dielectric substrate (;U÷" $, #R Fò C ’ Fò G astically compress RF wavelength, allowing resonant GPS (1575.42 MHz) and S-Band antennas to fit into compact CubeSat faces with Right-Hand Circular Polarization (RHCP).



ESP32 Dual-Core Type-C DevKit · Hardware & GPIO Avionics Bus

1. High-Dielectric Permittivity Miniaturization

The physical resonant length of a patch antenna depends inversely on the square root of the substrate's relative permittivity (;U÷"“

L "H c / (2 · f_r · " ;UöVfb•

In free air (;U÷" Ò ’Â SsRãC" Ô‡¢ u 2 çFVææ pould be ~95 mm long—too large for a 10×10 cm CubeSat face! Using barium titanate ceramic with ;U÷" Ò #R &VGV6W2 F†R pavelength by " 25 = 5, shrinking the patch down to a compact 25 mm × 25 mm footprint.

Key Highlights: High-dielectric ceramic (;U÷ "O # cC " R" Wavelength compression: ,µoB O »_07 „";U÷ " R" 60× ?B #]s#]sB mm standard size

2. Right-Hand Circular Polarization (RHCP)

GPS satellites transmit Right-Hand Circularly Polarized electromagnetic waves. A ceramic patch antenna uses a corner-truncated geometry or dual orthogonal probe feeds to generate two perpendicular electric field components with a 90° phase shift. This prevents severe signal fading caused by satellite tumbling and atmospheric Faraday rotation.

Key Highlights: RHCP matched to GNSS satellites • Immune to 90° Faraday cross-polarization drop • Axial ratio < 3 dB at boresight

MATHEMATICAL FORMULATIONS & PHYSICS DERIVATIONS

Microstrip Patch Resonant Length Equation

```
L = c / (2 * f_r * sqrt(eps_eff)) - 2 * delta_L
```

Calculates the physical length of a microstrip patch antenna taking into account dielectric permittivity and fringing fields.

Voltage Standing Wave Ratio (VSWR) & Return Loss

```
VSWR = (1 + |Gamma|) / (1 - |Gamma|);  Return_Loss (dB) = -20 * log10(|Gamma|)
```

Quantifies impedance matching between antenna feed line (50 ohms) and the ceramic patch.

EXAMINER VIVA QUESTIONS & DEFENSE ANSWERS

Q: Why is ceramic used as the substrate material in GPS patch antennas?

A: Ceramic materials (like barium titanate) possess a very high dielectric constant (;U÷" $, 20 to 40). Because resonant wavelength is inversely proportional to " ;U÷ , the ceramic substrate shrinks the physical dimensions of the antenna down to ~25 mm, allowing it to easily fit on a CubeSat.

CUBESAT 1U PROTOTYPE FLIGHT ROLE:

Mounted flush on the exterior solar-panel face of the 1U CubeSat to capture GPS L1 navigation signals from the GPS constellation without interfering with deployer rail ejection.

NASA Open MCT (Mission Control Technologies)

Next-Generation Open-Source Mission Control Telemetry Visualization

CORE THESIS & DEFINITION:

NASA Open MCT is an open-source, web-based mission control framework developed by NASA Ames Research Center and JPL. It provides real-time telemetry plotting, telemetry dictionary parsing, time-conductor scrubbing, and customizable mission layouts for spacecraft operations.



NASA Open MCT (Mission Control Technologies) · Aerospace Technical Reference

1. NASA Ames & JPL Open Architecture

Developed to replace bulky legacy desktop consoles. Open MCT operates natively inside modern web browsers (Chrome/Firefox) using HTML5, WebSockets, and Canvas graphics. It has powered real missions including lunar rovers (VIPER), ISS research payloads, and academic CubeSats.

Key Highlights: Developed by NASA Ames & JPL • 100% Open source (Apache 2.0) • Browser-based client-server design

2. The Telemetry Dictionary

In Open MCT, all satellite telemetry is governed by a Telemetry Dictionary (JSON format). It formally defines each telemetry point (e.g. cubesat.alt, cubesat.vbat, cubesat.temp, cubesat.pitch), specifying data types, physical measurement units (meters, volts, °C), valid limits, and display formatting.

Key Highlights: Structured telemetry point taxonomy • Metadata dictionary schema • Decouples telemetry provider from visualization

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: What is NASA Open MCT?

Ans: NASA Open MCT (Open Mission Control Technologies) is an open-source, web-based visualization framework developed by NASA Ames and JPL for mission planning, real-time telemetry monitoring, and data analysis of spacecraft and rovers.

Q2: What is the purpose of the Telemetry Dictionary in Open MCT?

Ans: The Telemetry Dictionary defines the metadata for all satellite data streams: variable keys, human-readable labels, physical engineering units (V, °C, meters), and alarm limit thresholds (nominal, warning, critical).

Red=Critical), artificial horizon attitude dials, and 3D spacecraft wireframes.

ORIGINAL DEVELOPER	LICENSE	FRONTEND STACK	INTEGRATION
NASA Ames Research Center & NASA JPL	Open Source (Apache License 2.0)	Vue.js / JavaScript / HTML5 WebSockets	Node.js WebSocket Telemetry Provider

Node.js bridge.js (Serial-to-WebSocket Gateway)

The Mission Control Middleware Connecting Physical Hardware to Web Cockpits

CORE THESIS & DEFINITION:

node bridge.js is a lightweight Node.js middleware server that reads raw telemetry strings from the Ground Station ESP32 USB COM port using SerialPort, parses the JSON payload, and broadcasts it over WebSockets to NASA Open MCT and browser dashboards.



Blynk IoT Cloud Platform · Mobile Telemetry Dashboard & Live Gauges

1. Why a Serial Bridge is Essential

Web browsers run in security sandboxes and cannot natively connect to local operating system hardware serial ports (COM3 or /dev/ttyUSB0) without restricted browser permissions, nor can they fan out one serial port to multiple team viewers. node bridge.js opens the USB COM port once and broadcasts telemetry over standard WebSockets (ws://localhost:8080).

Key Highlights: Bypasses browser hardware sandboxes • Multi-client fan-out distribution • Zero-dependency on proprietary software

2. Serial Ingestion Pipeline

The script imports the serialport npm package and uses ReadlineParser to split raw incoming byte buffers by newline ('\n'). When the Ground Station ESP32 prints a JSON telemetry line, the parser instantly fires a data event with the complete string.

Key Highlights: serialport package at 115200 baud • ReadlineParser newline buffering • Automatic USB disconnect/reconnect handling

EXAMINER VIVA QUESTIONS & TECHNICAL DEFENSE

Q1: Why is node bridge.js necessary in the ground station software stack?

Ans: Because web browsers cannot easily or reliably share a single raw hardware USB COM port across multiple applications or clients. node bridge.js handles hardware serial communication at 115200 baud and converts it into a high-speed WebSocket feed accessible by NASA Open MCT and web dashboards.

Q2: How does bridge.js broadcast telemetry packets to multiple viewers?

Ans: It runs a WebSocket server (typically on port 8080). When a complete telemetry line arrives from the serial port, it iterates through all connected client sockets (wss.clients.forEach) and transmits the JSON payload in real time.

RUNTIME	SERIAL PACKAGE	BAUD RATE	WEBSOCKET SERVER
Node.js (v18, v20, v22 LTS)	**serialport + @serialport/parser-readline**	**115200 bps (8-N-1)**	**ws on ws://localhost:8080**